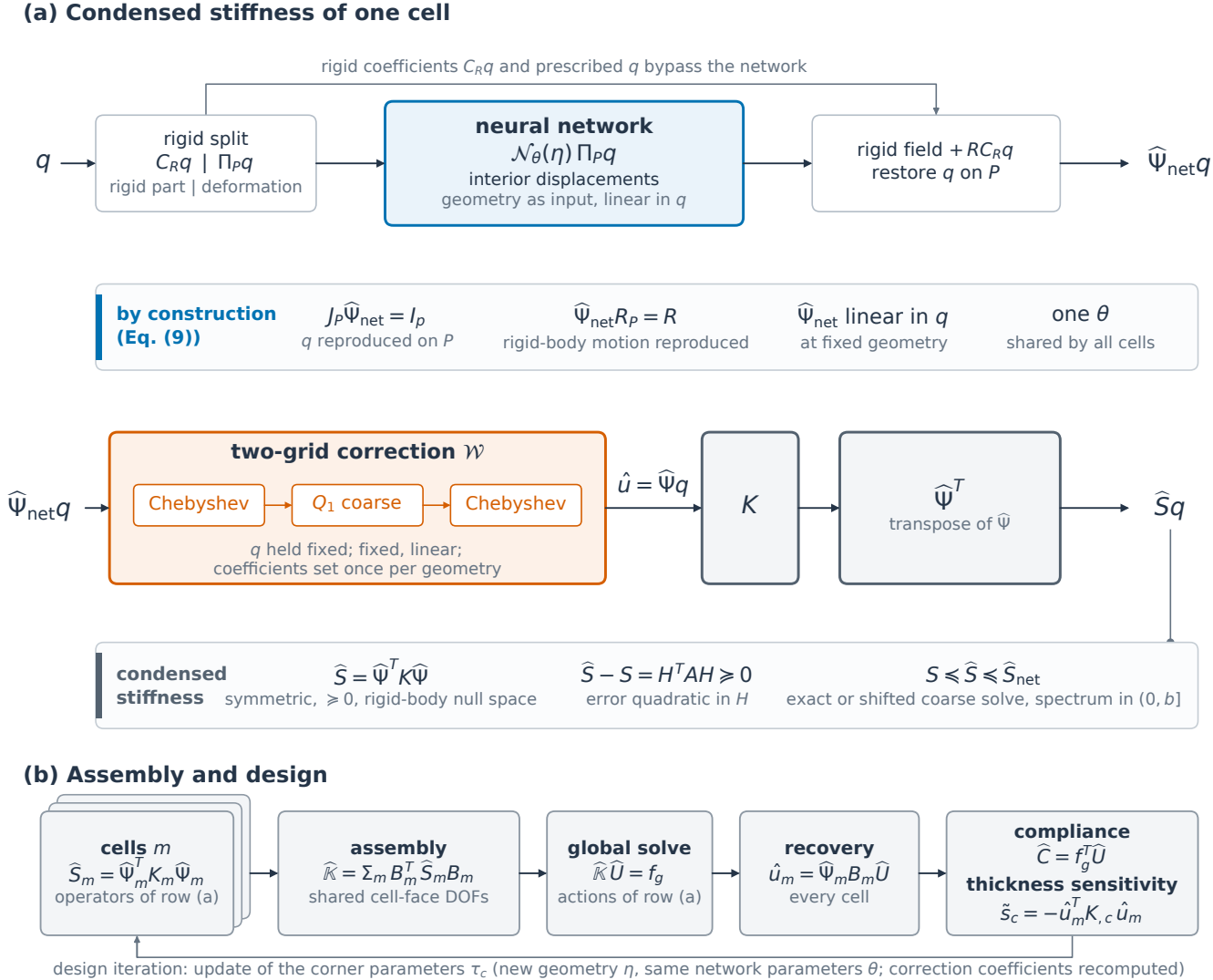
(a) Condensed stiffness of one cell
rigid coefficients $C_R q$ and prescribed q bypass the network
q
rigid split
$C_R q$ | $\Pi_P q$
rigid part | deformation
neural network
$\mathcal{N}_\theta(\eta)\,\Pi_P q$
interior displacements
geometry as input, linear in q
rigid field + RC_Rq
restore q on P
$\widehat{\Psi}_{\text{net}} q$
by construction (Eq. (9))
$J_P\widehat{\Psi}_{\text{net}} = I_p$
q reproduced on P
$\widehat{\Psi}_{\text{net}} R_P = R$
rigid-body motion reproduced
$\widehat{\Psi}_{\text{net}}$ linear in q
at fixed geometry
one θ
shared by all cells
$\widehat{\Psi}_{\text{net}} q$
two-grid correction $\mathcal{W}$
Chebyshev
Q_1 coarse
Chebyshev
q held fixed; fixed, linear;
coefficients set once per geometry
$\hat{u} = \widehat{\Psi} q$
K
$\widehat{\Psi}^T$
transpose of $\widehat{\Psi}$
$\widehat{S} q$
condensed stiffness
$\widehat{S} = \widehat{\Psi}^T K \widehat{\Psi}$
symmetric, $\succcurlyeq 0$, rigid-body null space
$\widehat{S} - S = H^T A H \succcurlyeq 0$
error quadratic in H
$S \preccurlyeq \widehat{S} \preccurlyeq \widehat{S}_{\text{net}}$
exact or shifted coarse solve, spectrum in $(0, b]$
(b) Assembly and design
cells m
$\widehat{S}_m = \widehat{\Psi}_m^T K_m \widehat{\Psi}_m$
operators of row (a)
assembly
$\widehat{\mathbb{K}} = \Sigma_m B_m^T \widehat{S}_m B_m$
shared cell-face DOFs
global solve
$\widehat{\mathbb{K}}\widehat{U} = f_g$
actions of row (a)
recovery
$\hat{u}_m = \widehat{\Psi}_m B_m \widehat{U}$
every cell
compliance
$\widehat{C} = f_g^T \widehat{U}$
thickness sensitivity
$\tilde{s}_c = -\hat{u}_m^T K_{,c}\, \hat{u}_m$
design iteration: update of the corner parameters τ_c (new geometry η, same network parameters θ; correction coefficients recomputed)